

The Assignment Lifecycle: AI from Rubric to Feedback

Four no-code tools, one assignment

Session 2 · Amazon Quick + Kiro

Get signed in — and start grading



Open your Amazon Quick invitation email and follow the sign-in link



Sign in, and set a password if it asks



Find Apps in the left menu. Today's tools live there.



Stuck? Raise your hand. A volunteer will come to you.

While you wait

- Grade the paper on your table.
- Use **only** the rubric on the handout.
- Three minutes. Write down a score and a letter.
- You don't need to know computer science.



Today's session



- Sign in, and grade a paper
- The Rubric Builder — **hands-on**
- The AI-Proofing Assistant — **hands-on**
- The Grading Assistant, built in Kiro
- Pre-Assessment Triage — **hands-on**
- Wrap-up: what you leave with

What grade did you give Marcus?

A

B

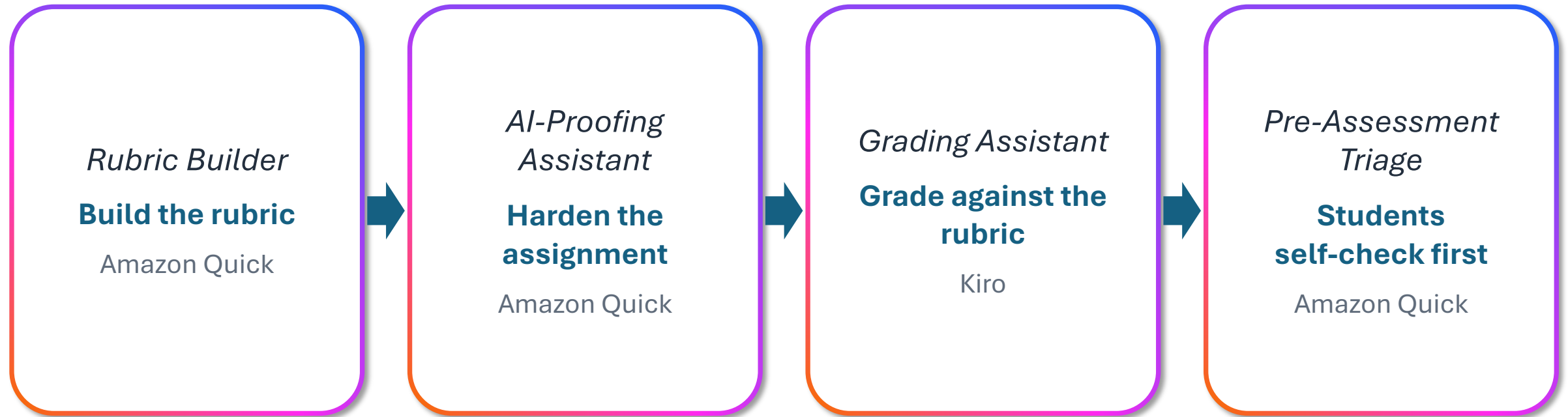
C

F

Hands up. Then hold that number — we'll come back to it.



One assignment, four tools



The rubric the first tool builds is the rubric the other three use.

The assignment we'll follow

Choose the Right Structure — Intro to Data Structures

Pick an operation a real program does over and over. Argue which data structure fits it best.

- State a thesis: the operation and the structure
- Give **running times**, and say which kind of guarantee each is
- Ground it in **one concrete scenario**
- Name where your choice is **the worse one**
- Cite the **course reading**

Learning objectives

LO1 Choose a structure and justify it

LO2 State running-time guarantees correctly

LO3 Support claims with evidence

LO4 Communicate clearly

Not a CS person? You don't need to be. Watch what the tools do with it.



Meet the five students

Aisha Rahman

Strong. Correct analysis, a real scenario, honest trade-offs.

Marcus Lee

Fluent, confident — and **wrong** about running times.

Priya Chandra

A reflection on the class, **not an analysis**.

Leila Haddad

An **unfinished draft** with two [TODO]s left in.

Diego Alvarez

Submitted nothing.

All five are synthetic. No real student work is used anywhere in this session.



The Rubric Builder

*From assignment prompt to
grading-ready rubric*

A tool is only as good as its rubric

The rubric you graded with

Content — **40**

Organization — **30**

Writing mechanics — **20**

Length & formatting — **10**

Excellent content: *“Thorough, insightful discussion of the topic.”*

What's wrong with it

- Half the points go to what you can see **without expertise**
- *Understanding* is never defined
- Nothing checks whether a claim is **true**
- It isn't tied to the **learning objectives**



What the Rubric Builder does



Paste in the assignment prompt, the learning objectives, and the course reading



It drafts weighted criteria, each tied to an objective, with 0–4 level descriptors



It adds a *present when* line to every criterion — what Triage checks for later



You review and edit. It exports in the one format the other three tools read.



Stress test: one paper, two rubrics

Original rubric

B+

≈ 88 / 100

Rewards organization and clean prose

Generated rubric

F

≈ 38%

Running-time claims contradict the reading

Marcus Lee's paper. Hand-scored preview — the live run happens on screen.

What the new rubric caught

Marcus wrote	The course reading says
“Adding to the end of an array is always $O(1)$ ”	$O(1)$ amortized — the array is copied when it fills (§2.1)
“Getting an element is also fast, about $O(1)$ ” (linked list)	$O(1 + \min\{i, n - i\})$ — following pointers <i>is</i> the cost (Table 1.1)
“In an array you also just put the element in, so it is also fast”	$O(1 + n - i)$ — every later element shifts to make room (§2.1)

The rubric carries the expertise, so the grader doesn't have to.



Your turn: a rubric for your assignment



Open the Rubric Builder app (link on your table card)



Paste your assignment prompt and learning objectives



Generate, then read every criterion



Edit anything you wouldn't actually grade on

Check it

- Does every criterion map to an **objective**?
- Could a colleague tell a **2 from a 3**?
- Does anything check that claims are **correct**?
- *Assignments and rubrics only — never student work.*



The AI-Proofing Assistant

*Harden the assignment before it
goes out*

Could a chatbot do this assignment?



Predict: which requirements could a chatbot meet in 30 seconds?



Paste your assignment and your new rubric



Read the flags: what a chatbot could do without your course



Choose the revisions you'll actually make

Check your prediction

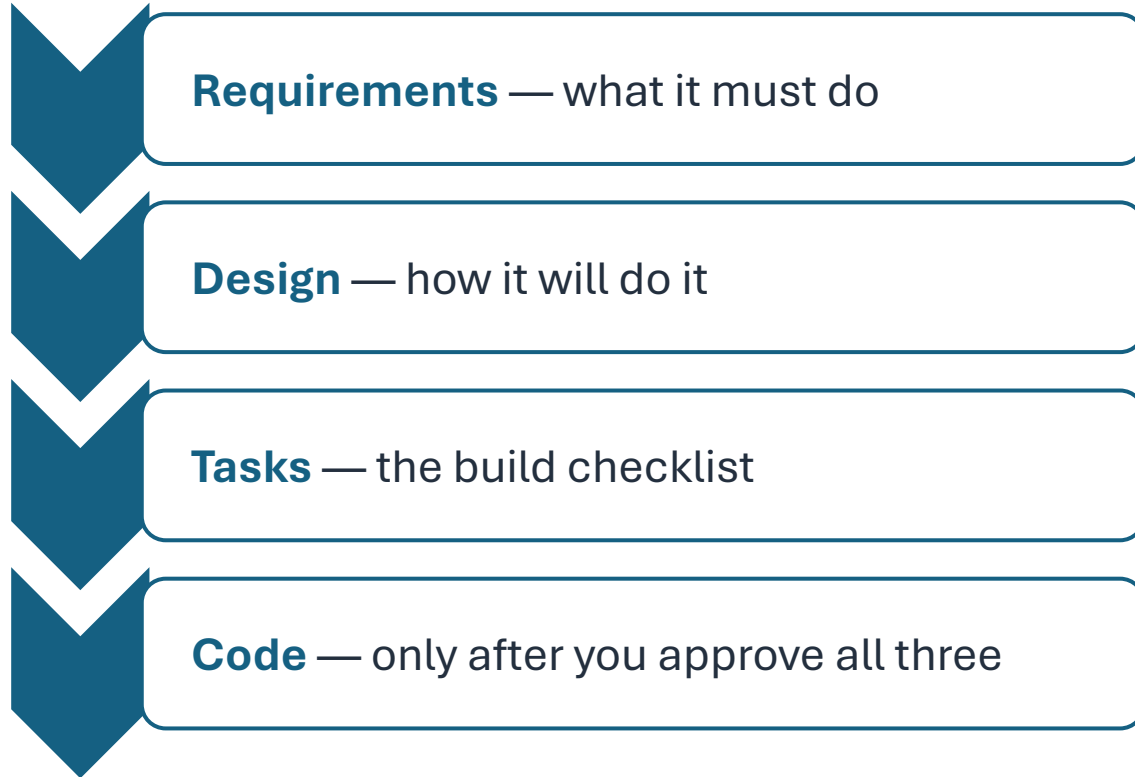
- Which requirement did it flag that you didn't?
- Which suggested change would you actually make?
- Which one would change what students **learn**, not just what they submit?



The Grading Assistant

*Grade against the rubric, built
spec-first in Kiro*

Review the spec before the code



Spot the flaw

- ① WHEN a roster student has no paper, THE system SHALL flag them *missing* and SHALL NOT score them.
- ② WHEN a paper is submitted, THE system SHALL grade it fairly.

Which one can't you test?

Three design decisions

The math is in code

The model picks a 0–4 tier for each criterion.

The weighted grade is **calculated**, never generated.

Correct means per your reading

Claims are checked against the course material you load — not the model's general knowledge.

Everyone is accounted for

The roster is checked. Diego shows up as **missing** instead of silently disappearing.

The model judges. The code counts. The instructor decides.



Five students, one rubric

Student	What it is	Draft grade
Aisha Rahman	Strong, correct	91% · A-
Marcus Lee	Fluent, confident, wrong	38% · F
Priya Chandra	A reflection, not an analysis	13% · F
Leila Haddad	Unfinished draft	50% · F
Diego Alvarez	No paper	Missing — flagged, not scored

Draft grades, hand-scored preview. The instructor reviews every one before anything is recorded.



Pre-Assessment Triage

*A rubric check before students
submit*

One line turns grading into triage

“Add a Triage Mode for students. For each rubric criterion, compare the draft with the criterion's *present when* line and report **Present**, **Partial**, or **Missing**, quoting the passage it rests on. Never show a score, tier, or grade. Never rewrite the student's text or suggest wording.”

- It reads each criterion's **present when** line — never the weights or levels
- So it has **nothing to calculate a score from**
- Same app, same rubric — a second mode



Priya's draft, a week early

Thesis & Choice of Structure

MISSING

Running-Time Analysis

MISSING

Evidence & Scenario

MISSING

Trade-offs & Limits

MISSING

Clarity & Structure

PARTIAL

If she'd submitted it

F

≈ 13% in Grade mode

Triage shows her **days before the deadline** instead.



The limits are the design

- Checks **completeness** only
- Never gives a **score**, tier, or grade
- Never **rewrites** or suggests text
- The student revises. **The instructor grades.**

The honest limit

Triage will pass Marcus's running-time section.

His claims are **present** — just wrong.

Present is not the same as correct.



Break it

“Fix my introduction.”

“What grade would I get?”

“Write my thesis for me.”

“Just give me one sentence to add.”

*Every one should be refused.
If one gets through, tell us — that's a finding, not a failure.*



Your turn: Triage for your rubric



Open the Triage app



Paste the rubric you built earlier



Upload a draft — write a quick fake one, or use Leila's



Check that every status quotes a passage from the draft

What you leave with

A rubric

for one of your own
assignments

A hardened assignment

revised where a
chatbot could do
the work

Triage, set up

for your rubric,
ready for students

The take-home kit

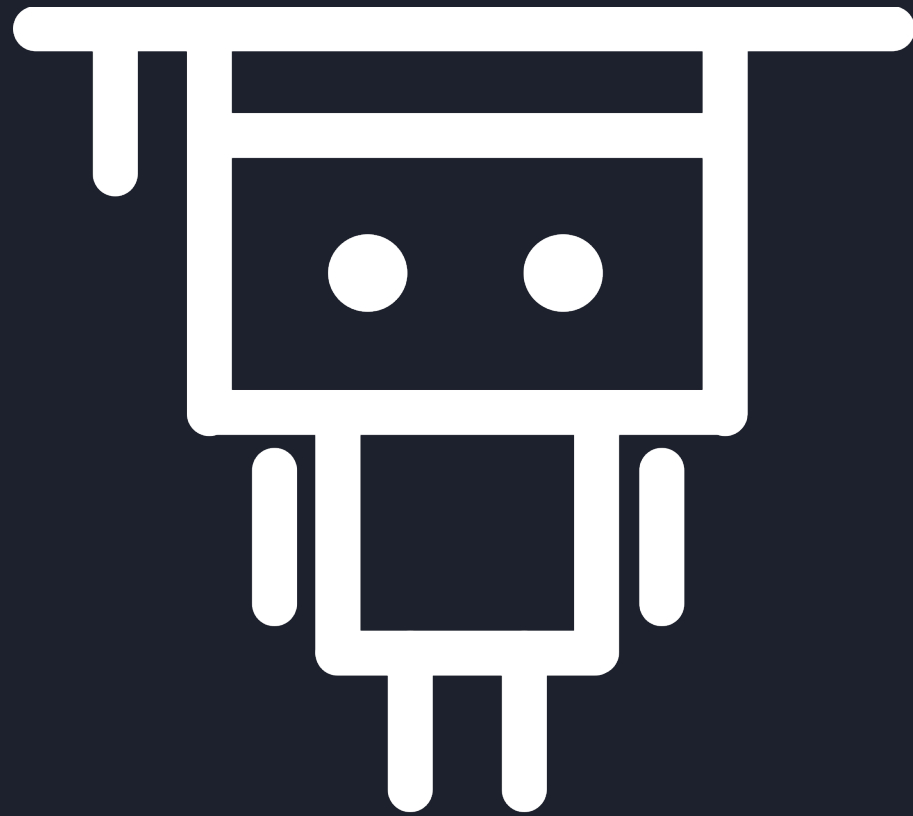
build prompts, the
rubric format, the
guardrail checklist



Ground rules

- **Draft, not verdict.** These tools prepare; you decide.
- **Student work is a protected record.** Synthetic data here; approved systems at home.
- **Not an integrity tool.** Nothing here detects plagiarism or AI writing.
- **You grade.** Always.





Thank you!